

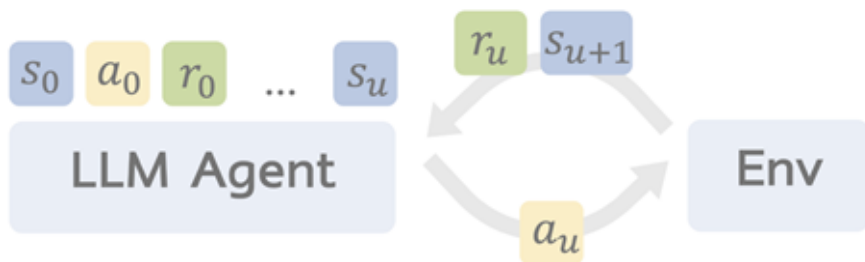
Federated Agent Reinforcement Learning

Canyu Chen*, Kangyu Zhu*, Zhaorun Chen, Zhanhui Zhou,
Shizhe Diao, Yiping Lu, Tian Li, Manling Li, Dawn Song
(*equal contribution)

AI Agents are Everywhere



Training AI Agents



Agent RL Loop

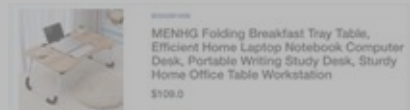


Web Agent



Embodied Agent

Typically requires **centralized** access to vast amounts of users' real-world task query and trajectory data, which are **inherently privacy-sensitive**.

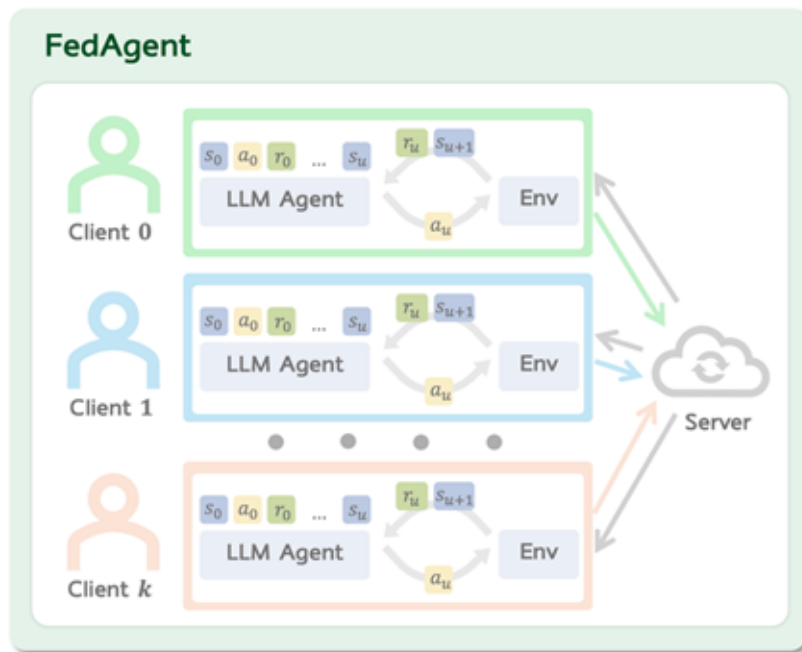


Can we train AI agents while protecting users' data privacy?

Embodied Agent

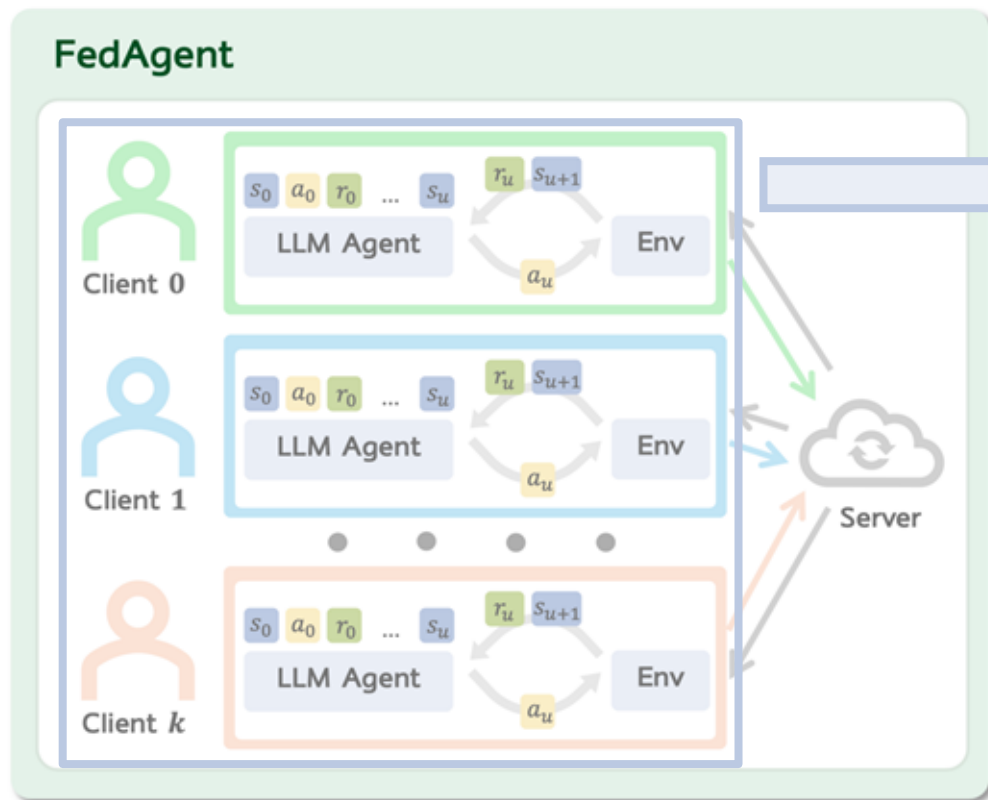
Typically requires *centralized* access to vast amounts of users' real-world task query and trajectory data, which are **inherently privacy-sensitive**.

FedAgent: Federated Agent Reinforcement Learning



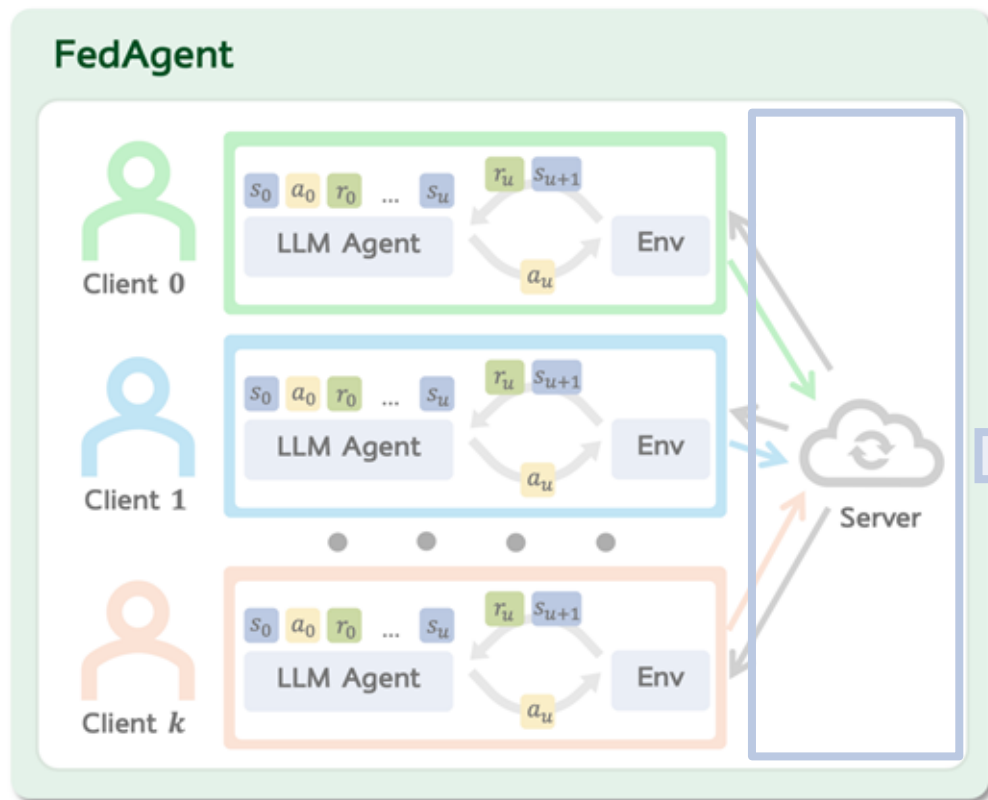
FedAgent enables **collaborative agent learning** across distributed clients **without sharing local data**, which preserve users' data privacy.

FedAgent: Federated Agent Reinforcement Learning



In each round, the current model is distributed to selected clients, who then train locally on their own data and send back their updated models.

FedAgent: Federated Agent Reinforcement Learning



The server aggregates these updates by averaging them to create an improved global model for the next round.

Challenges..

Conventional Federated RL typically assumes **low-dimensional** state and action spaces, **simple** tasks and environment interactions.

FedAgent involves **natural language** state and action spaces, **diverse** task formulations, and **complex** environment interactions.

Challenges..

Conventional Federated RL typically assumes **low-dimensional** state and action spaces, **simple** tasks and environment interactions.

FedAgent involves **natural language** state and action spaces, **diverse** task formulations, and **complex** environment interactions.

Is FedAgent Still Effective?

Convergence of FedAgent

Theorem 1 (Convergence of FEDAGENT). *Under Assumptions 1–5, suppose that at each communication round t the server uniformly samples without replacement a subset $S_t \subset [K]$ of size $M \leq K$ and aggregates only those clients' updates: $\theta_{t+1} = \theta_t + \frac{1}{M} \sum_{k \in S_t} \Delta \theta_{k,t}$, with the same local inner loop and notation as in Algorithm 1. Let each selected client perform τ local steps with stepsize η . Choose the stepsize $\eta = \frac{1}{L\tau}$ and let $\tilde{\theta}$ be a uniform random iterate drawn from $\{\theta_t\}_{t=0}^{T-1}$. Then*

$$\mathbb{E}[J(\theta^*) - J(\tilde{\theta})] \leq \frac{L}{\mu T} (J(\theta^*) - J(\theta_0)) + \frac{1}{2\mu} \left[\frac{G^2 + \sigma^2}{M} + \frac{2(K-M)}{M(K-1)} \zeta^2 + \frac{(\tau-1)^2}{2\tau^2} (G^2 + \sigma^2) \right].$$

FedAgent converges to a **neighborhood of the optimum** under the PL and smoothness conditions.

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: qwen2.5-1.5b/-3b/-7b, llama3.2-3b

Application Datasets:



MENHO Folding Breakfast Tray Table,
Efficient Home Laptop Notebook Computer
Desk, Portable Writing Study Desk, Study
Home Office Table Workstation
\$108.9



KPSP Folding Study Desk Bed Breakfast
Serving Tray Table Efficient Home Laptop
Notebook Computer Desk Portable
Study Desk for Small Space Bedroom

WebShop



ALFWorld

Decentralized Settings: samples per client,
clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity,
Coverage Heterogeneity, Hardness Heterogeneity

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: qwen2.5-1.5b/-3b/-7b, llama3.2-3b

Application Datasets:



MENHO Folding Breakfast Tray Table, Efficient Home Laptop Notebook Computer Desk, Portable Writing Study Desk, Sturdy Home Office Table Workstation
\$108.9



KPSP Folding Study Desk Bed Breakfast Serving Tray Table Efficient Home Laptop Notebook Computer Desk Portable
Alternative Desks for Small Space Bedroom

WebShop



ALFWorld

Decentralized Settings: samples per client, clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity, Coverage Heterogeneity, Hardness Heterogeneity



LLM Agents from **different families** and with **different scales**.

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: qwen2.5-1.5b/-3b/-7b, llama3.2-3b

Application Datasets:



MENHO Folding Breakfast Tray Table, Efficient Home Laptop Notebook Computer Desk, Portable Writing Study Desk, Sturdy Home Office Table Workstation

\$108.9



KPSP Folding Study Desk Bed Breakfast Serving Tray Table Efficient Home Laptop Notebook Computer Desk Portable

Alternative Desks for Small Spaces Bedroom

WebShop



ALFWorld

Decentralized Settings: samples per client, clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity, Coverage Heterogeneity, Hardness Heterogeneity



1. Real-world scenarios with **essential data privacy concerns**.
2. Require **complex reasoning process** and **multi-step environment interactions**.

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: qwen2.5-1.5b/-3b/-7b, llama3.2-3b

Application Datasets:



WebShop
MENHO Folding Breakfast Tray Table, Efficient Home Laptop Notebook Computer Desk, Portable Writing Study Desk, Sturdy Home Office Table Workstation
\$108.9



ALFWorld
KPSP Folding Study Desk Bed Breakfast Serving Tray Table Efficient Home Laptop Notebook Computer Desk Portable
Reversible Desk for Small Space Bedroom



ALFWorld

Decentralized Settings: samples per client, clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity, Coverage Heterogeneity, Hardness Heterogeneity

The number of samples per client: the sampling scope for each LLM agent's exploration.

The number of clients selected per communication round: controlling both computational parallelism and heterogeneity.

The number of local training batches per client per round: the extent of local optimization

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: qwen2.5-1.5b/-3b/-7b, llama3.2-3b

Application Datasets:



MENHO Folding Breakfast Tray Table, Efficient Home Laptop Notebook Computer Desk, Portable Writing Study Desk, Sturdy Home Office Table Workstation \$108.9



KPSIP Folding Study Desk Bed Breakfast Serving Tray Table Efficient Home Laptop Notebook Computer Desk Portable Roundtable Desk for Small Room Bedroom

WebShop



ALFWorld

Decentralized Settings: samples per client, clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity, Coverage Heterogeneity, Hardness Heterogeneity

Preference Heterogeneity: Clients Have Different Task Preferences.

Coverage Heterogeneity: Clients Have Different Task Sampling Scopes.

Hardness Heterogeneity: Clients Face Different Task Difficulties.

FedAgentGym: A Decentralized Agent Learning Environment

FedAgentGym

LLM Agents: gwen2.5-1.5b/-3b/-7b, llama3.2-3b

How to study the impact of each form of heterogeneity on FedAgent ***controllably***?

Decentralized Settings: samples per client, clients selected per round, epochs per client per round

Client Heterogeneity: Preference Heterogeneity, Coverage Heterogeneity, Hardness Heterogeneity



Have Different Task Sampling Scopes.

Hardness Heterogeneity: Clients Face Different Task Difficulties.

PreferencePartition

Algorithm 2 PREFERENCEPARTITION

Require: Category pools $\{\mathcal{I}_c\}_{c=1}^C$ with sizes n_c ; total clients K ; per-client set size L ; jitter ω

Ensure: Client datasets $X_1, \dots, X_K$ with $|X_k| = L$

- 1: $p_c \leftarrow n_c / \sum_{j=1}^C n_j$; $\ell_c \leftarrow \log \frac{p_c}{1-p_c}$ $\triangleright$ global mix + logit anchors
 - 2: **for** $k = 1$ to K **do**
 - 3: $z_c \sim \mathcal{N}(\ell_c, \omega^2)$ for $c = 1 \dots C$; $q_c \leftarrow \exp(z_c) / \sum_j \exp(z_j)$ $\triangleright$ larger $\omega \Rightarrow$ higher variance
 - 4: $(a_1, \dots, a_C) \sim \text{Multinomial}(L; q_1, \dots, q_C)$ $\triangleright$ category counts for client k
 - 5: **if** any $a_c > n_c$ **then** set $a_c \leftarrow \min(a_c, n_c)$ and redistribute leftover by q to classes with spare capacity $\triangleright$ capacity fix within a set
 - 6: $X_k \leftarrow \bigcup_{c=1}^C \text{SAMPLEWITHOUTREPLACEMENT}(\mathcal{I}_c, a_c)$
 - 7: **end for**
 - 8: **return** $\{X_k\}_{k=1}^K$
-

PreferencePartition: Based on **Multinomial Sampling**. One Hyperparameter ω can control the extent of **Preference Heterogeneity** across clients, while keeping the other heterogeneities **unchanged**.

PreferencePartition

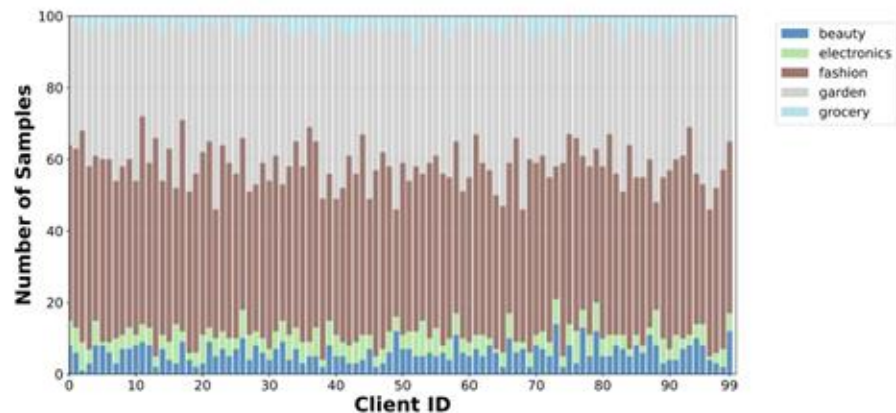


Figure 5: Client Distribution under Preference Heterogeneity (WebShop, $\omega = 0.1$).

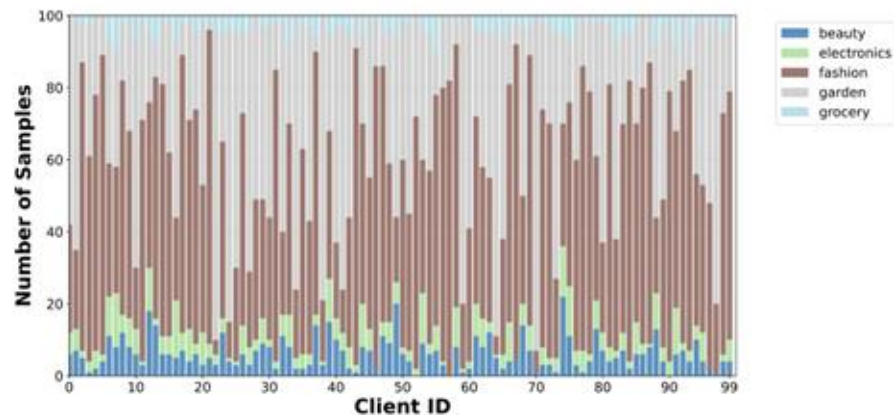


Figure 6: Client Distribution under Preference Heterogeneity (WebShop, $\omega = 0.9$).

CoveragePartition

Algorithm 2 PREFERENCEPARTITION

Require: Category pools $\{\mathcal{I}_c\}_{c=1}^C$ with sizes n_c ; total clients K ; per-client set size L ; jitter ω

Ensure: Client datasets $X_1, \dots, X_K$ with $|X_k| = L$

- 1: $p_c \leftarrow n_c / \sum_{j=1}^C n_j$; $\ell_c \leftarrow \log \frac{p_c}{1-p_c}$ ▷ global mix + logit anchors
 - 2: **for** $k = 1$ to K **do**
 - 3: $z_c \sim \mathcal{N}(\ell_c, \omega^2)$ for $c = 1 \dots C$; $q_c \leftarrow \exp(z_c) / \sum_j \exp(z_j)$ ▷ larger $\omega \Rightarrow$ higher variance
 - 4: $(a_1, \dots, a_C) \sim \text{Multinomial}(L; q_1, \dots, q_C)$ ▷ category counts for client k
 - 5: **if** any $a_c > n_c$ **then** set $a_c \leftarrow \min(a_c, n_c)$ and redistribute leftover by q to classes with spare capacity ▷ capacity fix within a set
 - 6: $X_k \leftarrow \bigcup_{c=1}^C \text{SAMPLEWITHOUTREPLACEMENT}(\mathcal{I}_c, a_c)$
 - 7: **end for**
 - 8: **return** $\{X_k\}_{k=1}^K$
-

CoveragePartition: Based on **Beta Distribution**. One Hyperparameter ξ can control the extent of **Coverage Heterogeneity** across clients, while keeping the others **unchanged**.

CoveragePartition

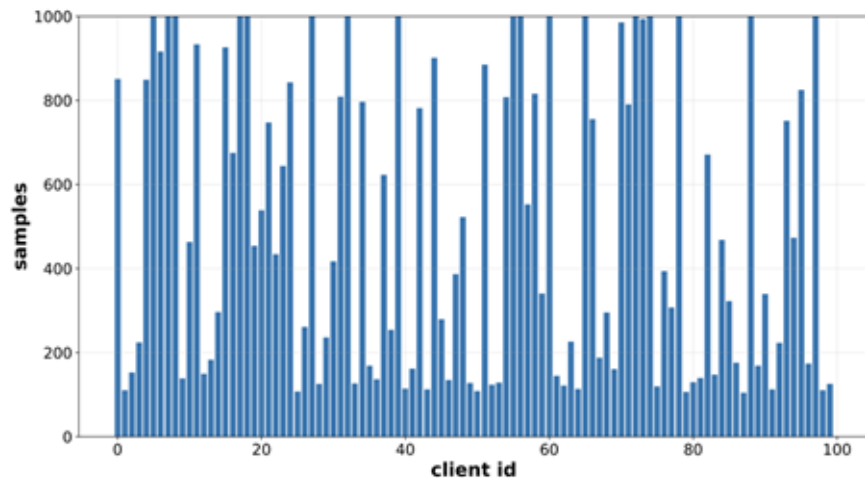


Figure 9: Client Distribution under Coverage Heterogeneity (WebShop, $\xi = 1$).

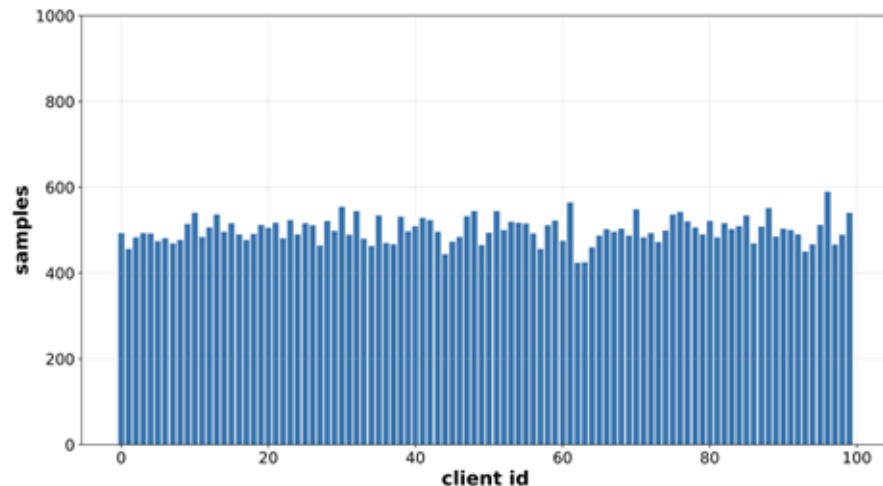


Figure 10: Client Distribution under Coverage Heterogeneity (WebShop, $\xi = 256$).

HardnessPartition

Algorithm 4 HARDNESSPARTITION

Require: total items N (indexed $1:N$); disjoint index sets $\mathcal{S}$ (successful) and $\mathcal{U}$ (unsuccessful) with $\mathcal{S} \cup \mathcal{U} = \{1:N\}$; total clients K ; per-client set size L ; Hyperparameters for COVERAGEPARTITION: bounds (ℓ, c, h) with $h \leq L$, dispersion ξ' , overlap r

Ensure: client datasets $X_1, \dots, X_K$ with $|X_k| = L$

- 1: $\{Y_k\}_{k=1}^K \leftarrow \text{COVERAGEPARTITION}(|\mathcal{S}|, K, (\ell, c, h), \xi', r)$ $\triangleright$ larger $\xi' \Rightarrow$ lower variance
 - 2: **for** $k = 1$ to K **do**
 - 3: $m_k \leftarrow L - |Y_k|$; $F_k \leftarrow \text{SAMPLEWITHOUTREPLACEMENT}(\mathcal{U}, m_k)$
 - 4: $X_k \leftarrow Y_k \cup F_k$
 - 5: **end for**
 - 6: **return** $\{X_k\}_{k=1}^K$
-

HardnessPartition: Based on **Beta Distribution**. One Hyperparameter ξ' can control the extent of **Hardness Heterogeneity** across clients, while keeping the others unchanged.

HardnessPartition

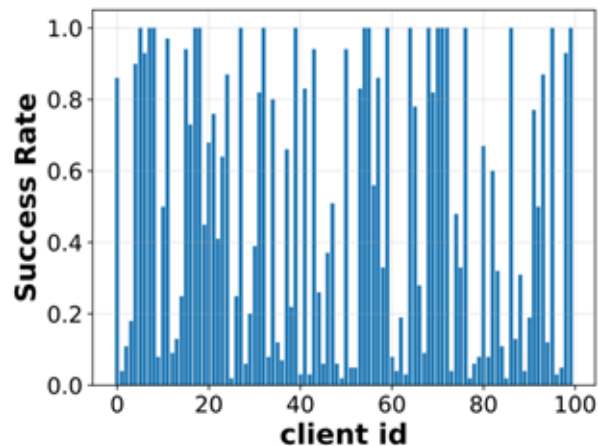


Figure 13: Client Distribution under Hardness Heterogeneity (WebShop, $\xi' = 1$).

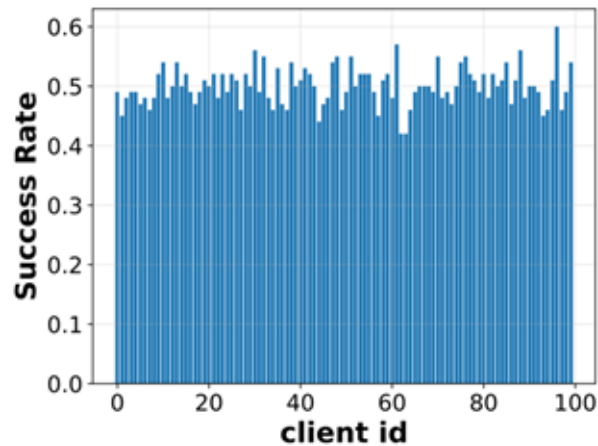


Figure 14: Client Distribution under Hardness Heterogeneity (WebShop, $\xi' = 256$).

Is FedAgent Comparable to Centralized Agent Learning?

What is the impact of Different Decentralized Settings?

What is the impact of Different Heterogeneity Challenges?

Is FedAgent Comparable to Centralized Agent Learning?

Method	ALFWorld							WebShop	
	Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
<i>Qwen2.5-1.5B-Instruct</i>									
Local (Client 21)	42.9	25.0	38.5	37.5	14.3	14.3	29.7	69.9	57.0
Local (Client 42)	50.0	37.5	76.9	25.0	42.9	14.3	45.3	75.1	53.1
Local (Client 84)	50.0	37.5	46.2	25.0	28.6	0.0	34.4	72.7	47.7
Centralized	64.3 $\pm$ 4.8	37.5 $\pm$ 0.9	69.2 $\pm$ 6.1	50.0 $\pm$ 2.2	42.9 $\pm$ 3.8	28.6 $\pm$ 0.4	51.6 $\pm$ 3.0	79.9 $\pm$ 4.7	57.8 $\pm$ 5.7
FEDAGENT	80.0 $\pm$ 4.2	75.0 $\pm$ 1.7	53.8 $\pm$ 4.3	37.5 $\pm$ 1.3	83.3 $\pm$ 4.7	50.0 $\pm$ 1.0	64.1 $\pm$ 2.8	83.2 $\pm$ 4.5	61.7 $\pm$ 1.8
<i>Qwen2.5-3B-Instruct</i>									
Local (Client 21)	41.5	12.5	34.9	51.0	18.9	21.2	31.3	59.8	55.0
Local (Client 42)	46.5	37.5	24.4	15.0	33.7	33.3	28.2	61.3	59.3
Local (Client 84)	22.8	27.5	39.1	46.3	48.3	36.5	29.9	77.6	58.6
Centralized	94.1 $\pm$ 0.9	80.0 $\pm$ 2.5	64.3 $\pm$ 1.4	42.9 $\pm$ 2.6	50.0 $\pm$ 2.7	22.2 $\pm$ 5.2	62.5 $\pm$ 4.2	70.0 $\pm$ 1.5	53.9 $\pm$ 2.8
FEDAGENT	95.5 $\pm$ 4.3	62.5 $\pm$ 3.0	49.7 $\pm$ 1.7	47.5 $\pm$ 2.4	85.3 $\pm$ 3.6	45.1 $\pm$ 2.1	65.2 $\pm$ 3.9	85.5 $\pm$ 3.4	63.1 $\pm$ 3.1
<i>Qwen2.5-7B-Instruct</i>									
Local (Client 21)	35.5	25.0	61.0	25.9	35.8	45.2	38.4	70.9	49.2
Local (Client 42)	29.0	45.0	18.8	25.6	15.9	38.0	42.1	78.2	33.6
Local (Client 84)	34.7	47.5	44.4	51.3	40.1	21.8	35.7	60.6	39.3
Centralized	93.7 $\pm$ 4.5	82.5 $\pm$ 2.1	71.5 $\pm$ 3.3	47.9 $\pm$ 3.7	63.2 $\pm$ 3.8	31.9 $\pm$ 1.0	73.3 $\pm$ 4.0	78.8 $\pm$ 2.8	64.7 $\pm$ 1.6
FEDAGENT	94.5 $\pm$ 2.3	85.0 $\pm$ 4.1	56.0 $\pm$ 0.8	62.5 $\pm$ 1.2	86.7 $\pm$ 2.9	42.8 $\pm$ 3.4	75.5 $\pm$ 2.9	89.0 $\pm$ 4.1	68.9 $\pm$ 3.8
<i>Llama-3.2-3B-Instruct</i>									
Local (Client 21)	39.8	50.0	17.9	40.0	20.7	34.0	38.1	65.3	50.5
Local (Client 42)	18.2	55.0	41.9	34.3	41.0	25.0	35.0	67.0	51.0
Local (Client 84)	29.9	32.5	39.0	18.9	18.8	37.6	29.7	70.2	55.7
Centralized	72.4 $\pm$ 4.6	62.5 $\pm$ 4.5	59.3 $\pm$ 3.1	45.2 $\pm$ 0.5	53.7 $\pm$ 2.2	27.9 $\pm$ 3.0	54.9 $\pm$ 2.9	72.3 $\pm$ 3.7	56.2 $\pm$ 1.6
FEDAGENT	83.7 $\pm$ 1.7	57.5 $\pm$ 6.0	60.6 $\pm$ 3.4	55.9 $\pm$ 0.9	65.3 $\pm$ 2.8	24.9 $\pm$ 3.1	61.2 $\pm$ 3.3	74.4 $\pm$ 4.9	57.8 $\pm$ 3.2

Is FedAgent Comparable to Centralized Agent Learning?

Method	ALFWorld							WebShop	
	Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
<i>Qwen2.5-1.5B-Instruct</i>									
Local (Client 21)	42.9	25.0	38.5	37.5	14.3	14.3	29.7	69.9	57.0
Local (Client 42)	50.0	37.5	76.9	25.0	42.9	14.3	45.3	75.1	53.1
Local (Client 84)	50.0	37.5	46.2	25.0	28.6	0.0	34.4	72.7	47.7
Centralized	64.3 $\pm$ 4.8	37.5 $\pm$ 0.9	69.2 $\pm$ 6.1	50.0 $\pm$ 2.2	42.9 $\pm$ 3.8	28.6 $\pm$ 0.4	51.6 $\pm$ 3.0	79.9 $\pm$ 4.7	57.8 $\pm$ 5.7
FEDAGENT	80.0 $\pm$ 4.2	75.0 $\pm$ 1.7	53.8 $\pm$ 4.3	37.5 $\pm$ 1.3	83.3 $\pm$ 4.7	50.0 $\pm$ 1.0	64.1 $\pm$ 2.8	83.2 $\pm$ 4.5	61.7 $\pm$ 1.8
<i>Qwen2.5-3B-Instruct</i>									
Local (Client 21)	41.5	12.5	34.9	51.0	18.9	21.2	31.3	59.8	55.0
Local (Client 42)	46.5	37.5	24.4	15.0	33.7	33.3	28.2	61.3	59.3
Local (Client 84)	22.8	27.5	39.1	46.3	48.3	36.5	29.9	77.6	58.6
Centralized	94.1 $\pm$ 0.9	80.0 $\pm$ 2.5	64.3 $\pm$ 1.4	42.9 $\pm$ 2.6	50.0 $\pm$ 2.7	22.2 $\pm$ 5.2	62.5 $\pm$ 4.2	70.0 $\pm$ 1.5	53.9 $\pm$ 2.8
FEDAGENT	95.5 $\pm$ 4.3	62.5 $\pm$ 3.0	49.7 $\pm$ 1.7	47.5 $\pm$ 2.4	85.3 $\pm$ 3.6	45.1 $\pm$ 2.1	65.2 $\pm$ 3.9	85.5 $\pm$ 3.4	63.1 $\pm$ 3.1
<i>Qwen2.5-7B-Instruct</i>									
Local (Client 21)	35.5	25.0	61.0	25.9	35.8	45.2	38.4	70.9	49.2
Local (Client 42)	29.0	45.0	18.8	25.6	15.9	38.0	42.1	78.2	33.6
Local (Client 84)	34.7	47.5	44.4	51.3	40.1	21.8	35.7	60.6	39.3
Centralized	93.7 $\pm$ 4.5	82.5 $\pm$ 2.1	71.5 $\pm$ 3.3	47.9 $\pm$ 3.7	63.2 $\pm$ 3.8	31.9 $\pm$ 1.0	73.3 $\pm$ 4.0	78.8 $\pm$ 2.8	64.7 $\pm$ 1.6
FEDAGENT	94.5 $\pm$ 2.3	85.0 $\pm$ 4.1	56.0 $\pm$ 0.8	62.5 $\pm$ 1.2	86.7 $\pm$ 2.9	42.8 $\pm$ 3.4	75.5 $\pm$ 2.9	89.0 $\pm$ 4.1	68.9 $\pm$ 3.8
<i>Llama-3.2-3B-Instruct</i>									
Local (Client 21)	39.8	50.0	17.9	40.0	20.7	34.0	38.1	65.3	50.5
Local (Client 42)	18.2	55.0	41.9	34.3	41.0	25.0	35.0	67.0	51.0
Local (Client 84)	29.9	32.5	39.0	18.9	18.8	37.6	29.7	70.2	55.7
Centralized	72.4 $\pm$ 4.6	62.5 $\pm$ 4.5	59.3 $\pm$ 3.1	45.2 $\pm$ 0.5	53.7 $\pm$ 2.2	27.9 $\pm$ 3.0	54.9 $\pm$ 2.9	72.3 $\pm$ 3.7	56.2 $\pm$ 1.6
FEDAGENT	83.7 $\pm$ 1.7	57.5 $\pm$ 6.0	60.6 $\pm$ 3.4	55.9 $\pm$ 0.9	65.3 $\pm$ 2.8	24.9 $\pm$ 3.1	61.2 $\pm$ 3.3	74.4 $\pm$ 4.9	57.8 $\pm$ 3.2

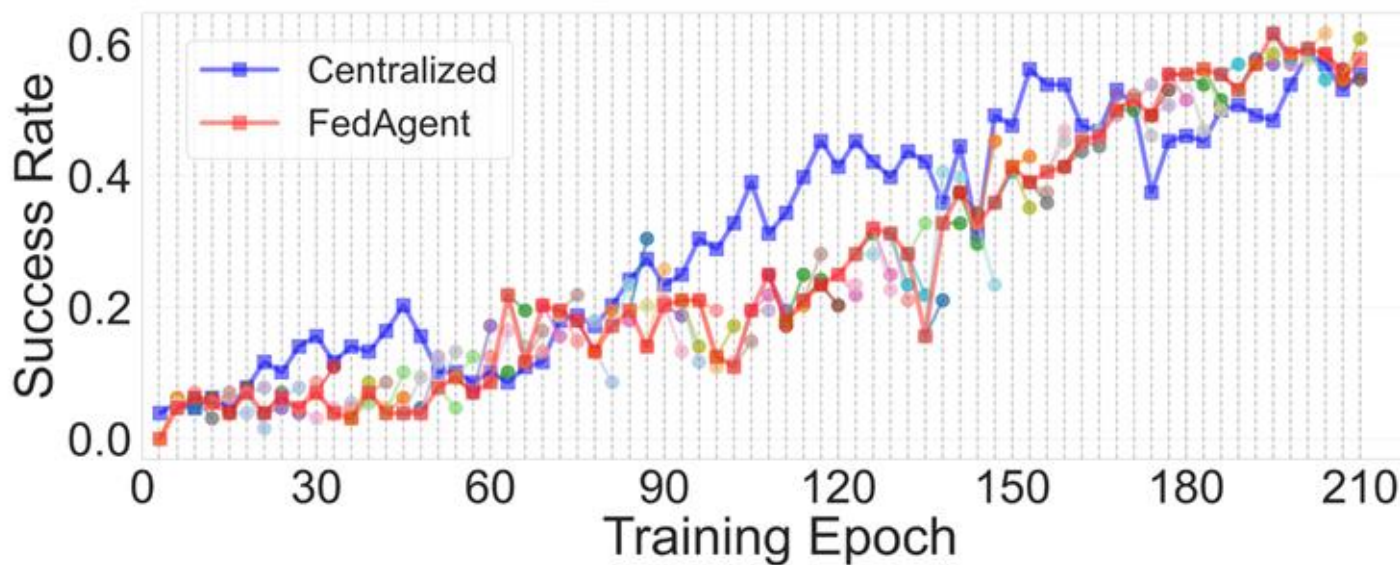
FedAgent achieves comparable performance to Centralized Agent Training.

Is FedAgent Comparable to Centralized Agent Learning?

Method	ALFWorld							WebShop	
	Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
<i>Owen2.5-1.5B-Instruct</i>									
Local (Client 21)	42.9	25.0	38.5	37.5	14.3	14.3	29.7	69.9	57.0
Local (Client 42)	50.0	37.5	76.9	25.0	42.9	14.3	45.3	75.1	53.1
Local (Client 84)	50.0	37.5	46.2	25.0	28.6	0.0	34.4	72.7	47.7
Centralized	64.3 $\pm$ 4.8	37.5 $\pm$ 0.9	69.2 $\pm$ 6.1	50.0 $\pm$ 2.2	42.9 $\pm$ 3.8	28.6 $\pm$ 0.4	51.6 $\pm$ 3.0	79.9 $\pm$ 4.7	57.8 $\pm$ 5.7
FEDAGENT	80.0 $\pm$ 4.2	75.0 $\pm$ 1.7	53.8 $\pm$ 4.3	37.5 $\pm$ 1.3	83.3 $\pm$ 4.7	50.0 $\pm$ 1.0	64.1 $\pm$ 2.8	83.2 $\pm$ 4.5	61.7 $\pm$ 1.8
<i>Owen2.5-3B-Instruct</i>									
Local (Client 21)	41.5	12.5	34.9	51.0	18.9	21.2	31.3	59.8	55.0
Local (Client 42)	46.5	37.5	24.4	15.0	33.7	33.3	28.2	61.3	59.3
Local (Client 84)	22.8	27.5	39.1	46.3	48.3	36.5	29.9	77.6	58.6
Centralized	94.1 $\pm$ 0.9	80.0 $\pm$ 2.5	64.3 $\pm$ 1.4	42.9 $\pm$ 2.6	50.0 $\pm$ 2.7	22.2 $\pm$ 5.2	62.5 $\pm$ 4.2	70.0 $\pm$ 1.5	53.9 $\pm$ 2.8
FEDAGENT	95.5 $\pm$ 4.3	62.5 $\pm$ 3.0	49.7 $\pm$ 1.7	47.5 $\pm$ 2.4	85.3 $\pm$ 3.6	45.1 $\pm$ 2.1	65.2 $\pm$ 3.9	85.5 $\pm$ 3.4	63.1 $\pm$ 3.1
<i>Owen2.5-7B-Instruct</i>									
Local (Client 21)	35.5	25.0	61.0	25.9	35.8	45.2	38.4	70.9	49.2
Local (Client 42)	29.0	45.0	18.8	25.6	15.9	38.0	42.1	78.2	33.6
Local (Client 84)	34.7	47.5	44.4	51.3	40.1	21.8	35.7	60.6	39.3
Centralized	93.7 $\pm$ 4.5	82.5 $\pm$ 2.1	71.5 $\pm$ 3.3	47.9 $\pm$ 3.7	63.2 $\pm$ 3.8	31.9 $\pm$ 1.0	73.3 $\pm$ 4.0	78.8 $\pm$ 2.8	64.7 $\pm$ 1.6
FEDAGENT	94.5 $\pm$ 2.3	85.0 $\pm$ 4.1	56.0 $\pm$ 0.8	62.5 $\pm$ 1.2	86.7 $\pm$ 2.9	42.8 $\pm$ 3.4	75.5 $\pm$ 2.9	89.0 $\pm$ 4.1	68.9 $\pm$ 3.8
<i>Llama-3.2-3B-Instruct</i>									
Local (Client 21)	39.8	50.0	17.9	40.0	20.7	34.0	38.1	65.3	50.5
Local (Client 42)	18.2	55.0	41.9	34.3	41.0	25.0	35.0	67.0	51.0
Local (Client 84)	29.9	32.5	39.0	18.9	18.8	37.6	29.7	70.2	55.7
Centralized	72.4 $\pm$ 4.6	62.5 $\pm$ 4.5	59.3 $\pm$ 3.1	45.2 $\pm$ 0.5	53.7 $\pm$ 2.2	27.9 $\pm$ 3.0	54.9 $\pm$ 2.9	72.3 $\pm$ 3.7	56.2 $\pm$ 1.6
FEDAGENT	83.7 $\pm$ 1.7	57.5 $\pm$ 6.0	60.6 $\pm$ 3.4	55.9 $\pm$ 0.9	65.3 $\pm$ 2.8	24.9 $\pm$ 3.1	61.2 $\pm$ 3.3	74.4 $\pm$ 4.9	57.8 $\pm$ 3.2

FedAgent consistently outperforms Local Agent Training.

Is FedAgent Comparable to Centralized Agent Learning?



Both paradigms (FedAgent and Centralized Agent Training) ultimately **converge to similar success rates** despite different training dynamics.

Is FedAgent Comparable to Centralized Agent Learning?



Takeway 1: FedAgent can be comparable to Centralized Agent Learning and beats Local Agent Learning.

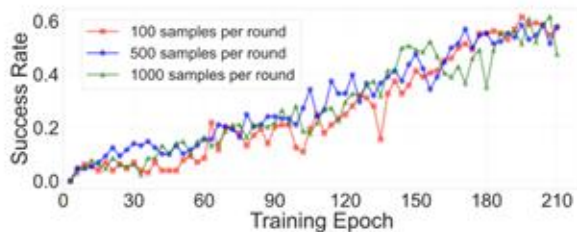
Both paradigms (FedAgent and Centralized Agent Training) ultimately **converge to similar success rates** despite different training dynamics.

Is FedAgent Comparable to Centralized Agent Learning?

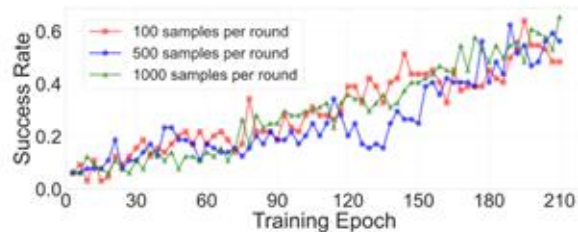
What is the impact of Different Decentralized Settings?

What is the impact of Different Heterogeneity Challenges?

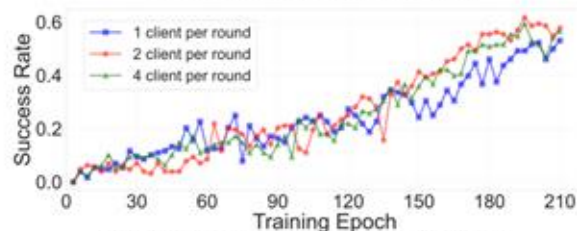
Impact of Decentralized Settings



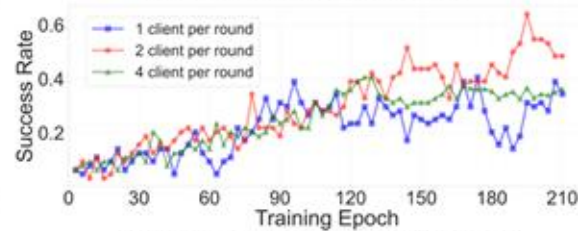
(a) Samples per client, WebShop



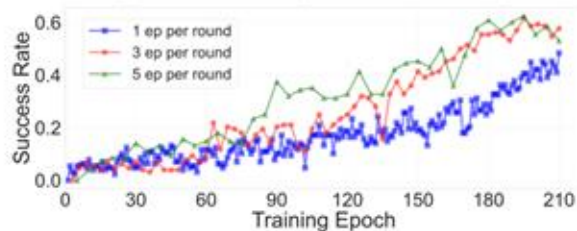
(b) Samples per client, ALFWorld



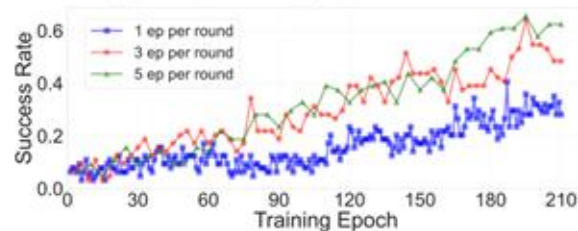
(c) Clients selected per round, WebShop



(d) Clients selected per round, ALFWorld

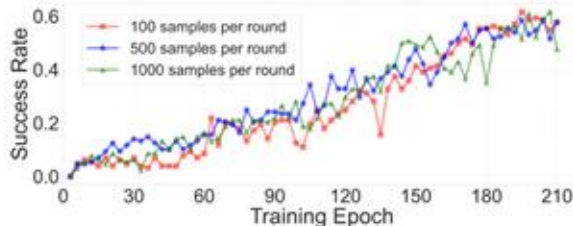


(e) Epochs per client per round, WebShop

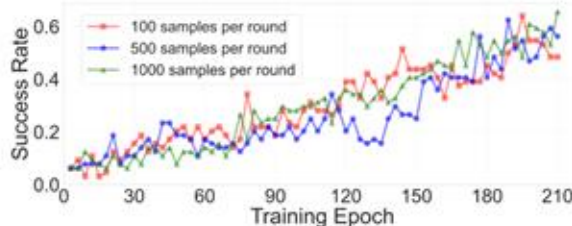


(f) Epochs per client per round, ALFWorld

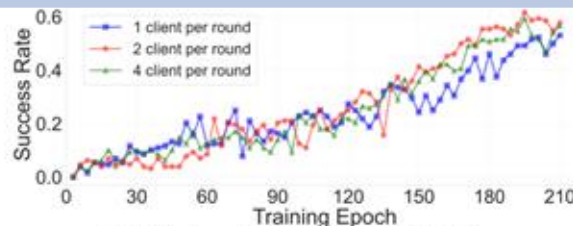
Impact of Decentralized Settings



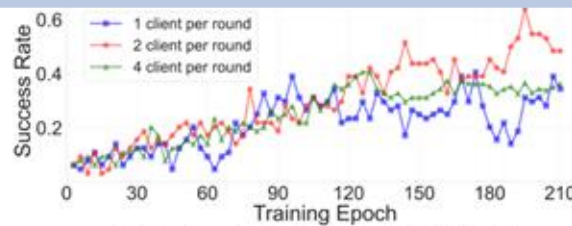
(a) Samples per client, WebShop



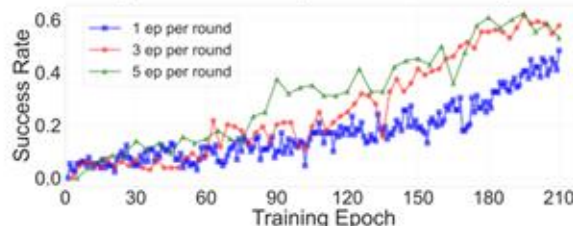
(b) Samples per client, ALFWorld



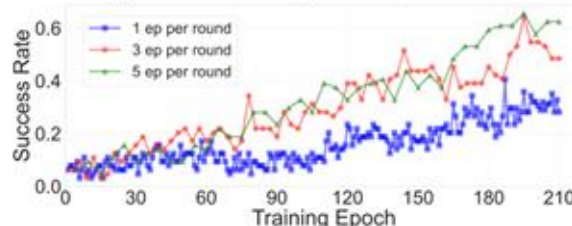
(c) Clients selected per round, WebShop



(d) Clients selected per round, ALFWorld



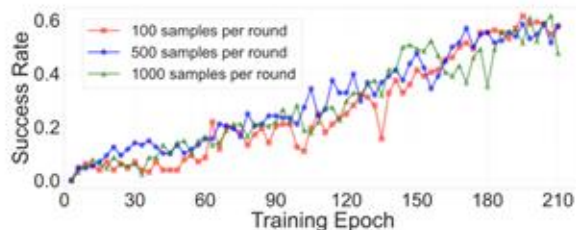
(e) Epochs per client per round, WebShop



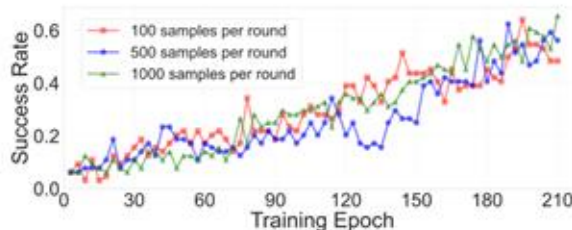
(f) Epochs per client per round, ALFWorld

FedAgent appears **insensitive** to the number of samples per client.

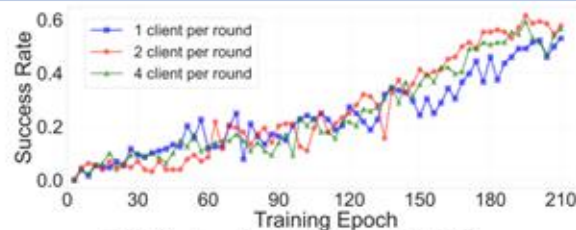
Impact of Decentralized Settings



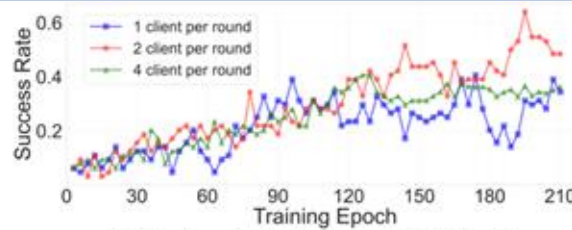
(a) Samples per client, WebShop



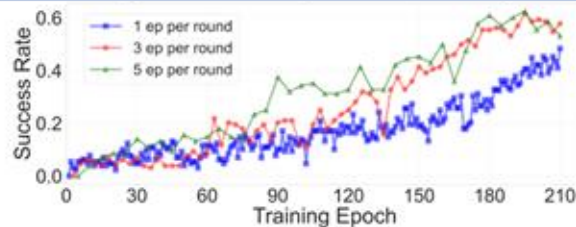
(b) Samples per client, ALFWorld



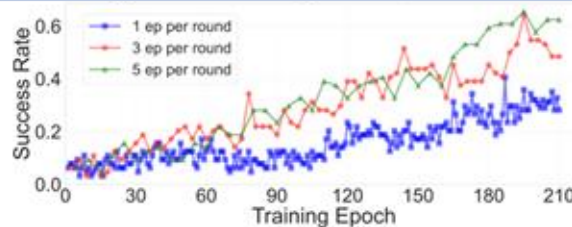
(c) Clients selected per round, WebShop



(d) Clients selected per round, ALFWorld



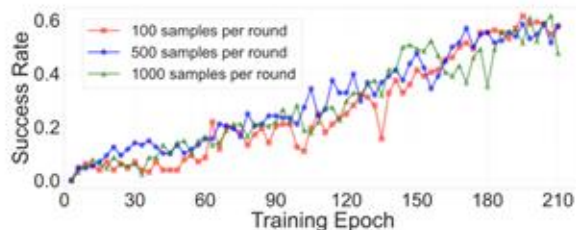
(e) Epochs per client per round, WebShop



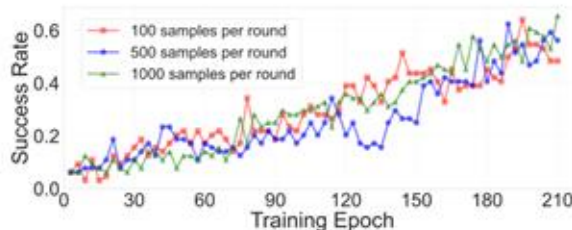
(f) Epochs per client per round, ALFWorld

FedAgent is **sensitive** to the number of clients selected per round on ALFWorld.

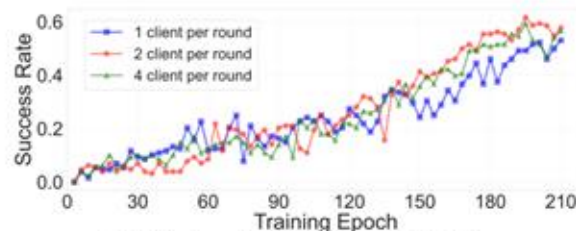
Impact of Decentralized Settings



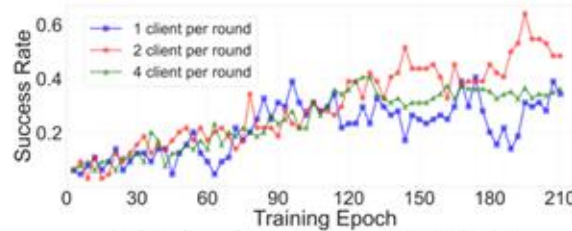
(a) Samples per client, WebShop



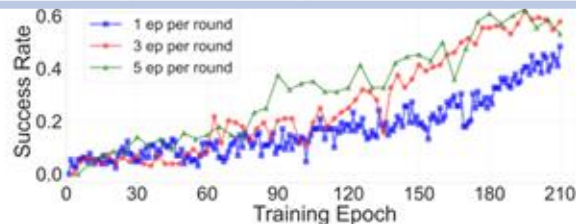
(b) Samples per client, ALFWorld



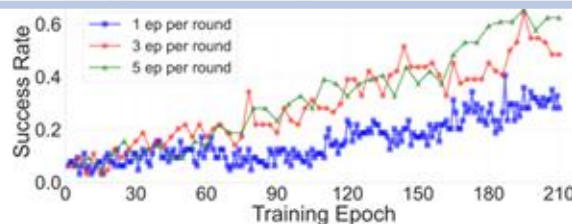
(c) Clients selected per round, WebShop



(d) Clients selected per round, ALFWorld



(e) Epochs per client per round, WebShop



(f) Epochs per client per round, ALFWorld

FedAgent is **sensitive** to the number of epochs per client per round on WebShop & ALFWorld.

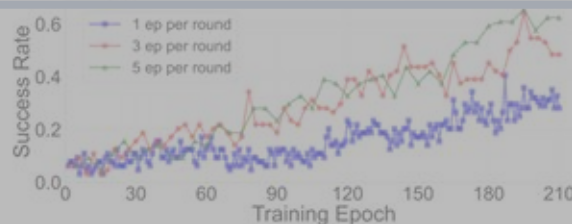
Impact of Decentralized Settings



Takeway 2: FedAgent's optimal decentralized configurations are environment-dependent.



(e) Epochs per client per round, WebShop



(f) Epochs per client per round, ALFWorld

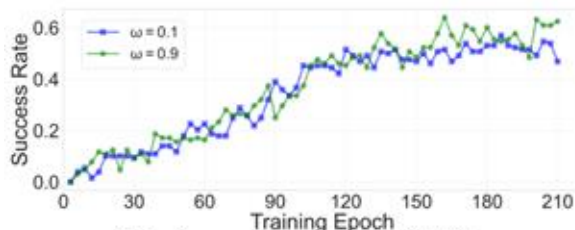
FedAgent is sensitive to the number of epochs per client per round on WebShop & ALFWorld.

Is FedAgent Comparable to Centralized Agent Learning?

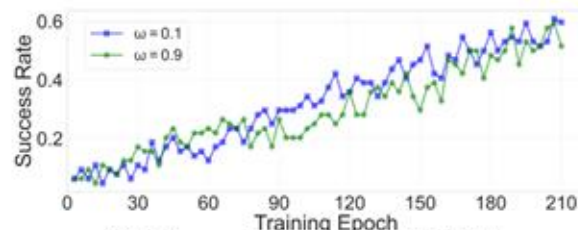
What is the impact of Different Decentralized Settings?

What is the impact of Different Heterogeneity Challenges?

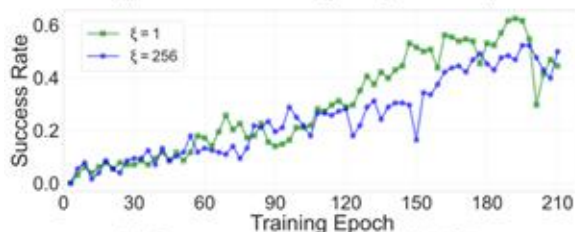
Impact of Heterogeneity Challenges



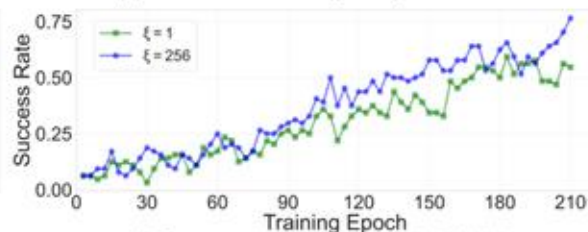
(a) Preference Heterogeneity, WebShop



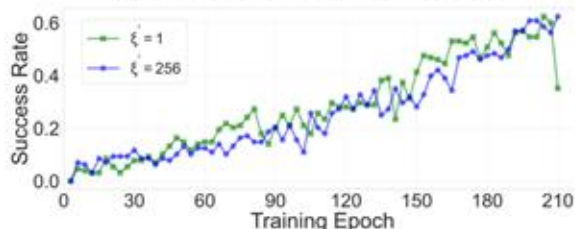
(b) Preference Heterogeneity, ALFWorld



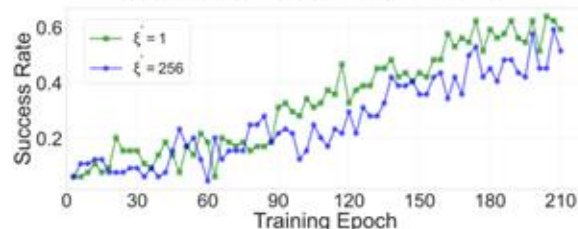
(c) Coverage Heterogeneity, WebShop



(d) Coverage Heterogeneity, ALFWorld

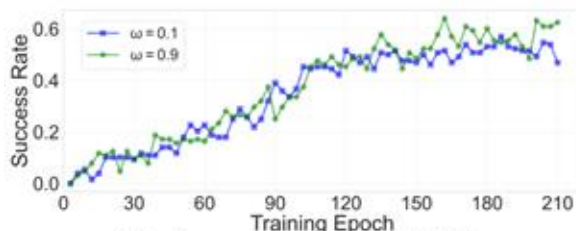


(e) Hardness Heterogeneity, WebShop

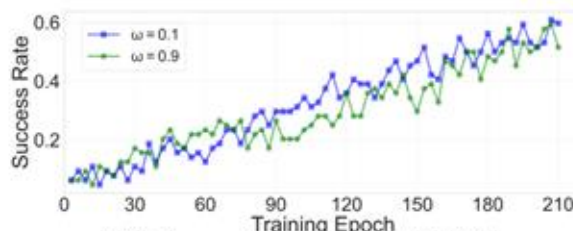


(f) Hardness Heterogeneity, ALFWorld

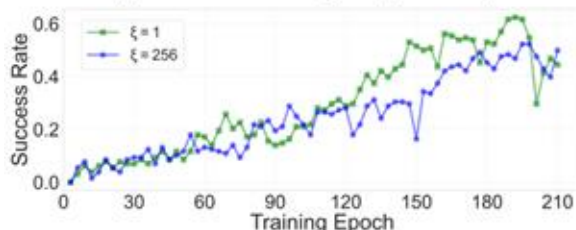
Impact of Heterogeneity Challenges



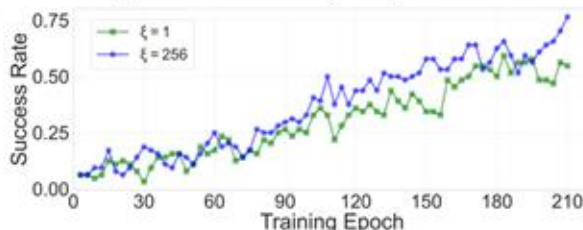
(a) Preference Heterogeneity, WebShop



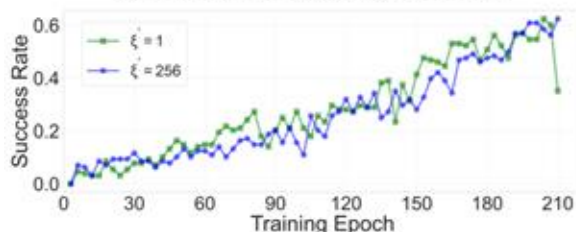
(b) Preference Heterogeneity, ALFWorld



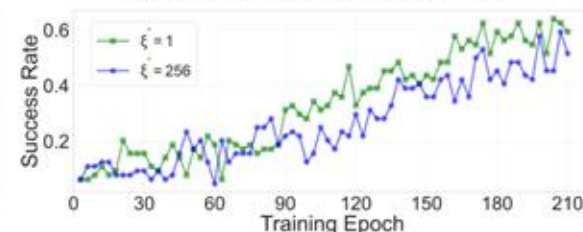
(c) Coverage Heterogeneity, WebShop



(d) Coverage Heterogeneity, ALFWorld



(e) Hardness Heterogeneity, WebShop

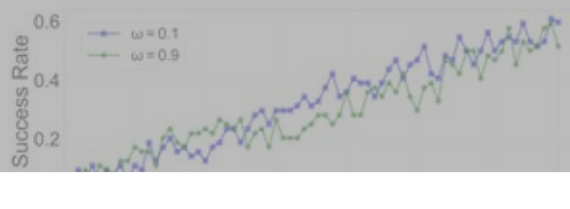


(f) Hardness Heterogeneity, ALFWorld

Green lines are high heterogeneity & Blue lines are low heterogeneity

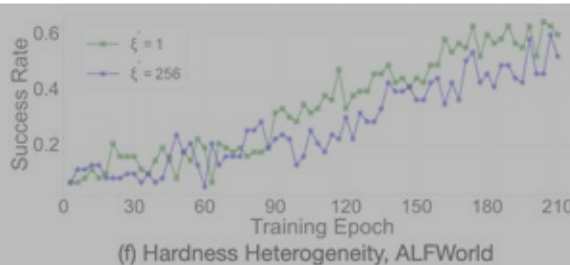
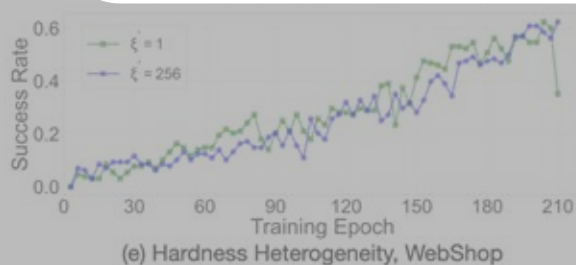
FedAgent ultimately **converge to similar success rates** across heterogeneities.

Impact of Heterogeneity Challenges



Green lines are high
heterogeneity. Blue lines

Takeway 3: FedAgent shows robustness against agent-specific heterogeneities.



converge to similar
success rates across
heterogeneities.

Thanks!